

CO3- AT3- Tree Traversal Challenges

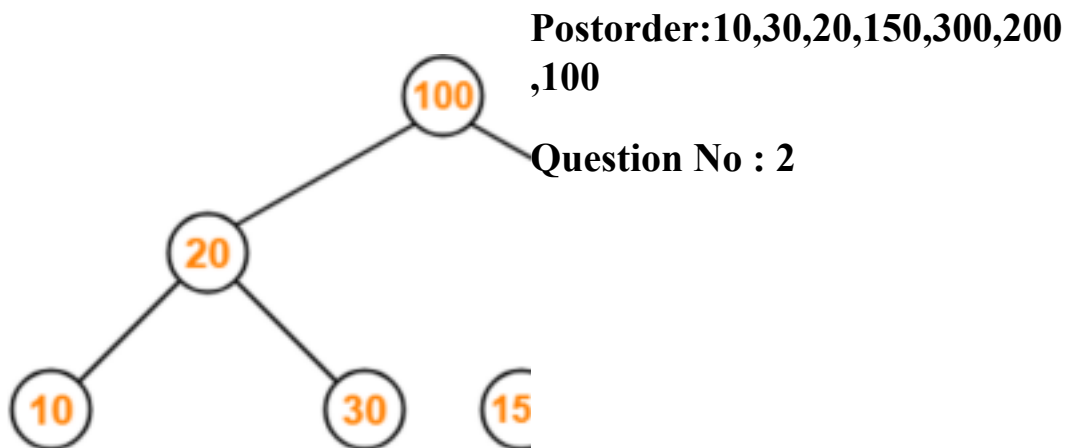
Write the traversal sequences for following trees

Name: H. Harshavardhan

Regno: 192511171

Sub: CSA0315 DATA STRUCTURE

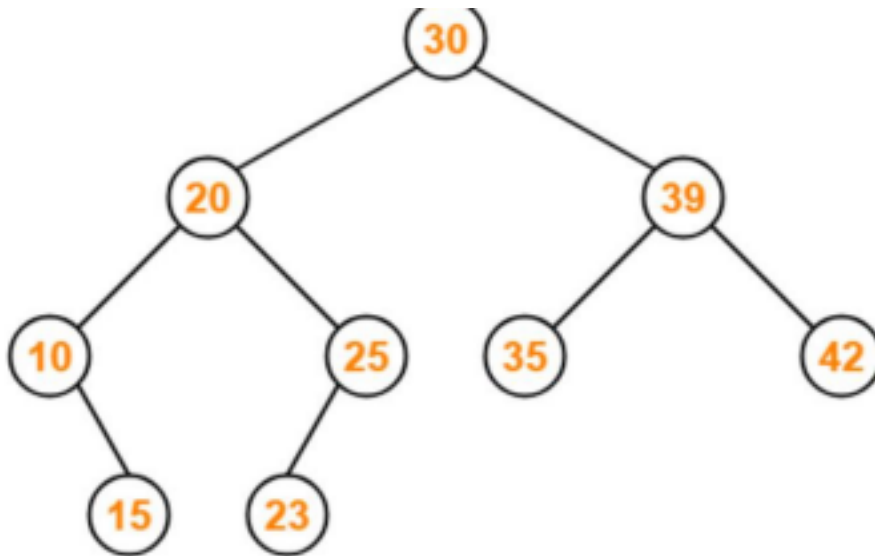
Question No : 1



Preorder: 100, 20, 10, 30, 200, 150, 300

Inorder: 10, 20, 30, 100, 150, 200, 300

Answer:



Preorder:30,20,10,15,25,23,39,35,42

Inorder:10,15,20,23,25,30,35,39,42

Postorder:15,10,23,25,20,35,42,39,30

Question No : 3

Assume the digits 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are added in that sequence into a binary search tree that is initially empty. The binary search tree follows the standard natural number ordering. What is the resulting tree's traversal sequence?

Inorder:0,1,2,3,4,5,6,7,8,9

Preorder:7,5,1,0,3,2,4,6,8,9

Postorder:0,2,4,3,1,6,5,9,8,7

Question No : 5

1

/

2 3

**/\ **

4 5 6

/

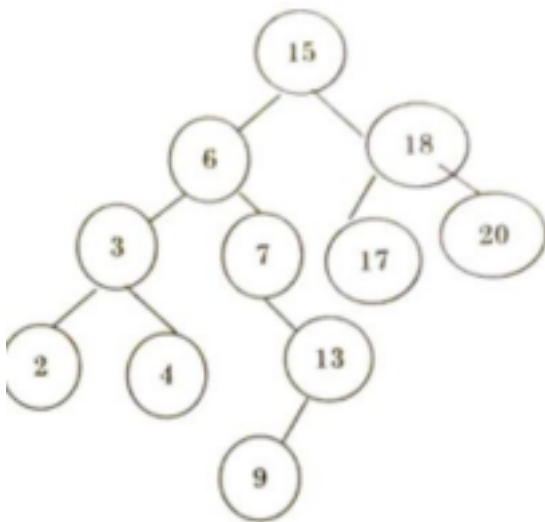
7

Preorder:1,2,4,7,5,3,6

Inorder:7,4,2,5,1,3,6

Postorder:7,4,5,2,6,3,1

Question No 6



Preorder:15,6,3,2,4,7,13,9,18,17,20

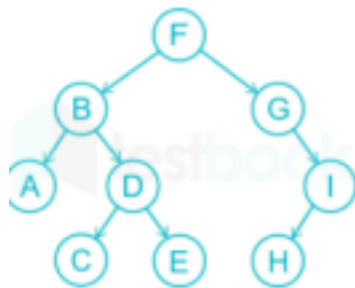
Inorder:2,3,4,6,7,9,13,15,17,18,20

Postorder:2,4,3,9,13,7,6,17,20,18,1

5

Question No 7

What is the sequence of nodes when applying in-order traversal on the binary tree given below?



1. A, B, C, D, E, F, I, H, G

2. A, C, D, E, B, F, G, H, I

3. A, B, C, D, E, F, G, H, I

4. I, H, G, F, E, D, C, B, A

Answer:

3

Question No 8

Consider the following Inorder and Preorder traversal of a tree

Inorder – D B E F A G H C

Preorder - A B D E F C G H

What is the Postorder Traversal of the same tree

1. D F E B H C G A

2. D F E B G H C A

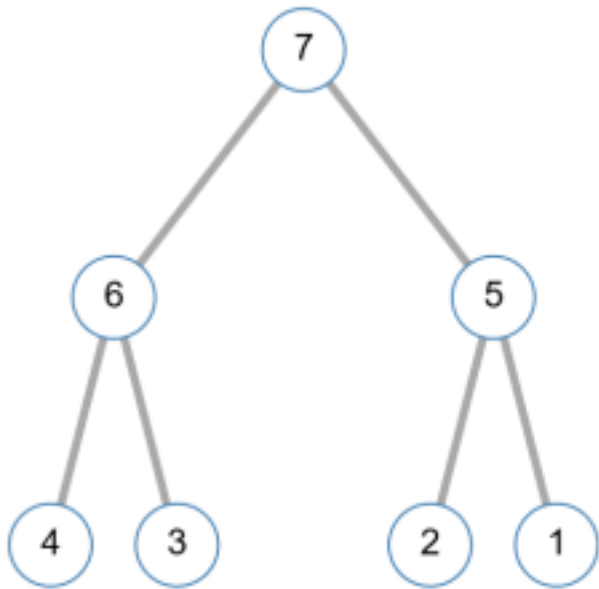
3. D F E B H G C A

4. D F E H B G C A

Answer:

3

Question No 9 : Find Inorder, Preorder and Postorder

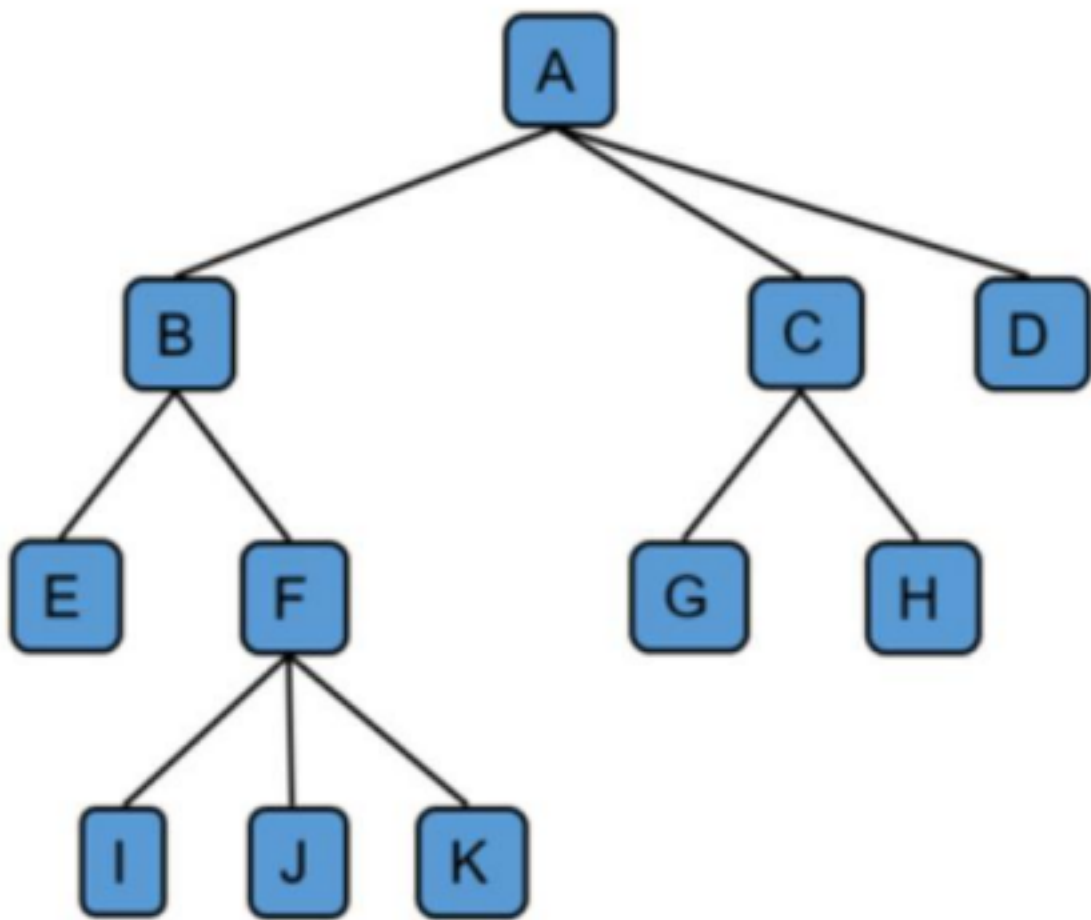


Preorder:7,6,4,3,5,2,1

Inorder:4,6,3,7,2,5,1

Postorder:4,3,6,2,1,5,7

Question No 10



Preorder: A,B,E,F,I,J,K,C,G,H,D

Inorder: A,B,C,D,E,F,G,H,I,J,K

Postorder: E,I,J,K,F,B,G,H,C,D,A